

1. Privacy-Constrained Deployment

1 point

A healthcare organization wants to use generative AI to summarize internal clinical notes for doctors. The data contains sensitive patient information and cannot leave the organization's infrastructure. Which approach is most appropriate?

- ☐ A large, general-purpose cloud-hosted model
- ☐ A creative model optimized for storytelling
- ☒ An open-source model deployed locally
- ☐ A high-temperature configuration of a frontier model

2. Specialization vs Scale

1 point

A software team wants AI assistance for code review, focusing on identifying security vulnerabilities and best practices in a specific programming language. Which option best fits this task?

- ☐ A model optimized for image generation
- ☐ A general-purpose model with a creative prompt
- ☐ A smaller model with no domain knowledge
- ☒ A fine-tuned or specialized model for code analysis

3. When One Model Isn't Enough

1 point

A company builds an AI assistant that answers customer questions, routes complex issues to experts, and generates internal reports. Which approach best reflects an effective system design?

- ☒ Use a routing or mixture-of-experts approach
- ☐ Use one large model for all tasks
- ☐ Switch models randomly to increase diversity
- ☐ Increase temperature for better performance

4. Cost-Aware Model Choice**1 point**

A startup needs to generate short, repetitive internal status summaries thousands of times per day. The task is low-risk and well-defined. Which model choice is most appropriate?

- ☐ The largest available frontier model
- ☐ A creative model with high temperature
- ☒ A specialized or distilled model
- ☐ A research-oriented reasoning model

5. Open vs Proprietary Tradeoff**1 point**

A research team wants full transparency into model behavior, reproducibility, and the ability to audit outputs over time. Which model strategy best supports this goal?

- ☐ Highly creative consumer-facing models
- ☐ Proprietary black-box models
- ☒ Open-source models with known architectures
- ☐ Models optimized for conversational fluency

6. Misuse of Model Choice**1 point**

A team switches to a larger and more expensive model to fix hallucinations in factual responses. Why is this approach likely flawed?

- ☒ Model size does not address grounding or verification
- ☐ Cost directly reduces accuracy
- ☐ Hallucinations are caused by temperature alone
- ☐ Larger models always hallucinate more

7. Hybrid System Design

1 point

An organization needs fast responses for common queries, deep reasoning for complex cases, and strict data governance. Which system design best fits?

- ☐ A single fine-tuned model for all tasks
- ☒ Multiple specialized models with routing
- ☐ A high-temperature creative model
- ☐ One general-purpose model

 You completed the Integrity Check for this submission. [View Integrity Check](#)

Your grade: 100%

Your latest: 100% • Your highest: 100% • To pass you need at least 80%. We keep your highest score.

Next item →

1. Privacy-Constrained Deployment

1 / 1 point

A healthcare organization wants to use generative AI to summarize internal clinical notes for doctors. The data contains sensitive patient information and cannot leave the organization’s infrastructure. Which approach is most appropriate?

✔ Nice work

2. Specialization vs Scale

1 / 1 point

A software team wants AI assistance for code review, focusing on identifying security vulnerabilities and best practices in a specific programming language. Which option best fits this task?

✔ Nice work

3. When One Model Isn’t Enough

1 / 1 point

A company builds an AI assistant that answers customer questions, routes complex issues to experts, and generates internal reports. Which approach best reflects an effective system design?

✔ Nice work

4. Cost-Aware Model Choice

1 / 1 point

A startup needs to generate short, repetitive internal status summaries thousands of times per day. The task is low-risk and well-defined. Which model choice is most appropriate?

✔ Nice work

5. Open vs Proprietary Tradeoff

1 / 1 point

A research team wants full transparency into model behavior, reproducibility, and the ability to audit outputs over time. Which model strategy best supports this goal?

✓ Nice work

6. Misuse of Model Choice

1 / 1 point

A team switches to a larger and more expensive model to fix hallucinations in factual responses. Why is this approach likely flawed?

✓ Nice work

7. Hybrid System Design

1 / 1 point

An organization needs fast responses for common queries, deep reasoning for complex cases, and strict data governance. Which system design best fits?

✓ Nice work